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Thermodynamics and equilibrium. A two-page handout for testing the Study Hub uploader.

Glossary

Enthalpy — the heat content of a system at constant pressure
Entropy : how spread out the energy of a system is
Gibbs free energy :: the energy usable for work at constant pressure
Catalyst - a substance that speeds a reaction without being consumed
Mole — mol
Entropy — second definition, ignored because the first one wins
Le Chatelier's principle — equilibrium shifts to oppose the disturbance
The second law of thermodynamics as stated in the course notes — total entropy of an isolated system never decreases

Two lines that will not become terms

Electronegativity
the tendency of an atom to attract shared electrons in a covalent bond

Copied out of a two-column table

Molar volume at STP	22.4 L/mol at 273 K and 1 atm

Worked example

The combustion of one mole of methane releases about 890 kJ of energy at 298 K.
A reaction whose Gibbs free energy change is negative is spontaneous at that temperature.
Raising the temperature by 10 K roughly doubles the rate of many room temperature reactions.
A catalyst lowers the activation energy, so more collisions carry enough energy to react.

A superscript, split onto its own row

One mole of any substance contains 6.022×10^{23} particles, which is Avogadro's number.